

SVI
SPECTRAVIDEO

Flight Simulator
user's manual

I N T R O D U C T I O N

The Spectravideo Flight Simulator has been designed as an entertainment programme to give the user relatively realistic control of a large four engined aircraft operating between selected airfields throughout Australasia.

The Flight simulator will run on an SV328 computer or on an SV318 which has a 16k ram expansion fitted. The programme is loaded from the tape using the RUN"fs" command. After a short time of loading the screen will display the title page while the loading of the main programme continues. On completion of the loading the programme will run automatically. This will take 9 minutes.

Complete control of the programme can be achieved at the keyboard, or if desired, one or two joysticks may be used for the principal aircraft controls.

Some of the main features and options incorporated within the simulator programme include:-

- a) The selection of a flight between any two of thirteen airports throughout Australasia. Including take-off and landing at the same airport for practice if desired.
- b) Practice landing routine into Auckland airport. Landings, particularly with approach under instrument display only, are the most difficult part of a flight and this routine has minimal effects of cross-wind to assist new users.
- c) Divert routine. During a flight this allows the selection of an alternative airport for landing. This option is useful if the initially selected fuel load proves to be inadequate and an alternative airport may be closer.
- d) Time Skip. During straight and level flight this option will shorten the actual time required to complete a flight normally of long duration.
- e) Map display. During the selection of a flight the map is displayed to show the actual locations of all the airfields and a summary of preflight conditions. During the flight the map may be displayed to give an accurate summary of flight conditions and to show the aircraft position and airfields on the map.

Relative locations of the airfields give the user the flexibility of selecting flights varying in duration from approximately 30 minutes to over 6 hours. Apart from take-off and landing all flying is by instrument guidance only.

For a completely new user with no prior knowledge of flying an aircraft there may appear a lot to learn, with many controls to master, many instruments to scan and reading to digest, all at once. However with a

INTRODUCTION cont'd.

period of practice and familiarisation, some patience, and the guidance of this instruction guide, any user should be able to take-off and fly the aircraft successfully and complete the final approach and landing. At worst, a flight ending in a disastrous crash on the simulator can damage nothing, with the possible exception being the users pride. Those users who are more familiar with flight theory are also recommended to read this guide to establish the aircraft performance criteria, the available options within the programme and how they may be used, and the aircraft and programme controls.

The following section of the manual shall cover all the necessary information to allow the Flight Simulator to be used fully and successfully, including aircraft controls and instruments, the programme options available, and some basic theory of flight. Section 3 provides a quick reference for the controls and performance criteria.

The programme has been carefully checked and run for many hours and to the best of the designer's knowledge, operates correctly with no bugs having become apparent. In the event of repeated difficulties in attempting to run the programme, it is suggested the user should have their tape cassette checked for correct loading, check the CAPS LOCK key is in the 'off' position, and read once again the relevant sections of this guide to ensure correct understanding of the programme and the controls involved.

It is the designer's hope that the Flight Simulator will provide some challenge and many hours of enjoyment to all users.

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SPECTRA VIDEO

FLIGHT SIMULATOR

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SECTION 1

In this, the main section of the manual, all the necessary information for flying with the Flight Simulator is presented, from the initial stage of flight selection through to final landing and shutdown. Some elementary details of flight theory and controls has been included at the start to provide some understanding of the behaviour and criteria of an aircraft in flight and the relevance of the various instruments and controls.

It must be emphasised that, while the information holds true in a real life flight situation, the programme is in no way intended to be a course of instruction for flying, or anything more than a source of entertainment. Any user wishing to pursue a formal training in flying should do so through the recognised organisations available.

Reference should be made to Section 3 for the relevant keyboard or joystick controls to achieve the required aircraft and programme controls.

1:1 BASIC FLIGHT THEORY & CONTROLS

Before proceeding with the Flight Simulator a brief outline of the basic principles of flight will be given, and we start by considering how the aircraft is kept aloft and under control using various control surfaces.

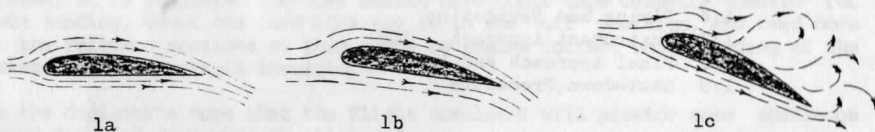


Figure 1

The required lift is produced by the airflow over the main wings which are shown in typical profile in figure 1. As the wing passes through the air the airstream is divided and deflected to flow both over and under the wing as shown in fig. 1a. The profile of the wing causes greater deflection over the top surface. With a longer path to travel over the wing and the corresponding greater air velocity, the result is a lesser air pressure above the wing and a greater pressure acting on the underside of the wing. This pressure differential will lift the wing providing the relative air velocity is great enough.

Lift can be increased by tilting the wing as shown in fig 1b, giving an increased angle of attack. As this angle increases, the airflow willact more directly on the underside of the wing, increasing the required deflection to the airflow around the wing and giving a greater drag factor. Hence, although lift will be increased with the greater angle of attack which will occur at take-off and during rapid climbing, more thrust and therefore more fuel will be required to maintain the aircraft velocity and the air velocity over the profile of the wing. The necessary angle of attack will be dependent on the speed of the aircraft through the air, the load to be carried and the rate of climb required etc.

If the angle of attack is increased excessively as shown if fig 1c, the drag on the wing will become too great and the airflow over the wing will become turbulent. At this point the lift applied to the wing will reduce dramatically with a resulting stall or loss of control and the aircraft will descend rapidly. A loss of lift and stall will also occur if the relative air velocity over the wing is allowed to reduce below a safe level.

SECTION 1:1 Cont'd.

Fitted within the trailing edge of the wings are adjustable sections called flaps. These may be lowered under the control of the pilot to various settings to increase the lift at low speed such as during take-off, final approach and landing. The increase in lift is quite significant but also however, is the increase in the drag. Therefore for efficiency and fuel economy the flaps are raised for normal flight once the the velocity of the aircraft is sufficient to maintain adequate lift on the wings. Figure 2 shows the profile of the flaps in the raised and lowered positions.

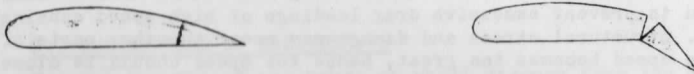


Figure 2

The final control surface to be considered on the main wings are the ailerons. These are also within the trailing edge of the wings but are located further out towards the ends of the wings. These ailerons also tilt but have two main differences when comparing them with the flaps, apart from the fact that they are normally smaller in size. The ailerons may tilt up from the normal zero setting as well as downwards and, they work or move in opposite directions. While the left aileron will tilt upwards when the control yoke is moved left, the right aileron will tilt down. This will reduce the lift on the left wing slightly and increase the lift on the right wing, causing the craft to roll to the left. Moving the control yoke to the right will cause a corresponding roll to the right. During turns to change direction the pilot will roll the aircraft inwards into the turn using the ailerons while using the rudder, controlled by foot pedals to turn the craft. The inwards banking of the aircraft will eliminate the sideslip which would occur if the rudder was to be used on its own.

For simplicity on the Flight Simulator programme, only one set of controls is required to give controlled turning of the aircraft without sideslip, both in the air and on the ground. The rudder may be considered as mechanically linked to the ailerons.

The elevators are the only remaining control surfaces of interest. These are located on the rear edges of the tailplane fins and behave in a similar manner to that described for the flaps and ailerons by deflecting the airflow off the rear of the tailplane. Pulling back on the control yoke will tilt the elevators up, causing the tail of the aircraft to dip. This gives an increase in the aircraft pitch or a nose-high attitude with a corresponding increase in the angle of attack of the main wings as mentioned earlier. Pushing the control yoke forwards will tilt the elevators down giving the opposite effect, lifting the tail of the aircraft.

All of the control surfaces which have been described in this section are dependent on the aircraft speed through the air being sufficient to ensure adequate air velocity over all the surfaces to give control. Loss of lift is therefore only one cause of stall, the other being loss of control in the air.

To complete this section brief comments will be made on the remaining controls or conditions of importance. Firstly thrust, the propulsion from the engines to overcome the drag and provide motion. In the Flight Simulator the

SECTION 1:1 Cont'd.

aircraft has four engines which may be started up or shut down independently. The throttle control is linked to all four engines to increase or decrease the power or thrust of the engines. On landing the throttle may be taken through its zero position to a negative value, providing reverse thrust to reduce the speed of the aircraft relatively rapidly until the conventional brakes may be used effectively.

The landing gear or undercarriage must be fully down and locked whenever the aircraft is on the ground. In flight the landing gear is retracted within compartments in the body of the craft to reduce drag and ensure maximum economy and efficiency, and to prevent excessive drag loadings at high speed causing damage to the gear. Structural stress and damage may occur to other parts of the aircraft if the speed becomes too great, hence the speed should be closely monitored to ensure it remains high enough to maintain control and prevent stall, and low enough to eliminate the risk of structural damage.

Many of the controls may initially appear slow in response. This is partly due to the limited time the programme spends monitoring input functions while continuously processing complex and repetitive calculations and display routines. However, this response is largely in keeping with the anticipated performance of a large aircraft which may weigh up to 400 tonnes when fully laden.

1:2 FLIGHT SELECTION

With the above limited theory in mind it is time to turn to the Simulator and complete the initial stage of flight selection. Assuming the programme has loaded correctly as outlined in the Introduction, the map should now be displayed on the screen.

Those wishing to use the Practice Landing routine should select this with the 'P' control and refer to Section 2:5 for further details.

To select a normal flight between any two of the listed airports, or even to take-off and land at the one airport, proceed as follows. Use the up or down arrow keys to select the airport which is to serve as the departure point. As the individual airports are highlighted in green on the list, the corresponding location will be indicated on the map. With the required departure airport shown, press the ENTER key to pass this information to the programme. Repeat this selection and confirming process for the destination airport.

Once the airports are confirmed the display will detail the distance, directional bearing, wind strength and direction, and runway directions or headings, and prompt the user to select the required fuel load for the flight. Key in the desired fuel load and again use the ENTER key to pass this to the programme. The fuel load may be any value from 10 to 150 tonnes. Typical fuel consumption is 4 to 5 tonnes per 100 miles of flight depending on wind strength and direction, and this itself may vary during the flight. The consequence of loading insufficient fuel are somewhat obvious but the selection of an excessive fuel load is also bad practice, with the greater load on take-off, the unnecessary load to be carried in flight causing extra consumption in itself, and a much more difficult landing with extra load and slightly impaired handling and response.

Before moving from the selection procedure, two observations and mental notes should be made. Firstly the direction of the runways, particularly the runway at the destination as this is not shown on the main flight instrument

SECTION 1:2 Cont'd.

display, but will be important during the initial landing approach as detailed later. Secondly, consider the strength and direction of the wind in relation to the departure runway to establish whether the aircraft will tend to drift towards the left or right side of the runway during take-off. If this drift is not compensated for the aircraft may actually crash off the side of the runway.

With the flight selection and fuel loading processes completed, pressing the ENTER key once again will change the display to the flight instrument panel in readiness for the start of the actual flight.

1:3 DESCRIPTION OF INSTRUMENTS

For the benefit of new users we shall digress from the actual flight briefly and outline the instruments on the display, some of which may initially appear rather obscure.

The clock at the top left of the display will start to run only when the brakes are released on take-off and will be stopped at the completion of the shut-down procedure. If the map is displayed during the flight the clock is stopped and all programme data including the aircraft position etc are suspended.

The flight wind indicator shows both the strength of the wind in knots, the strength will generally increase at higher altitudes, and the wind source direction e.g. from 180° is from the south, from 090° is from the east etc.

The brakes and gear indicators are self explanatory as is the flaps indicator. It should be noted that each control to raise or lower the flaps will operate in increments of 10° between fully raised 0° and maximum flaps down setting of 30° .

The indicators for the four engines simply show the status for each of the engines, green if running normally, red if shut-down or blue if shutting down or starting up.

The elevator and aileron indicators show the setting for the relevant control surfaces as previously described. The relative rate of turn is shown also on the aileron indicator. The pitch indicator shows the attitude of the aircraft, positive values of pitch designating nose up and negative values nose down

The grid at the top centre of the display and the three associated marker lamps will be explained in the section dealing with instrument approach.

In the centre of the display, during take-off and final approach and landing, is the runway indicator. This will show the side drift of the aircraft across the runway, the relative height of the aircraft above the runway surface and the length of runway used. After take-off this section of the display will change to the artificial horizon indicator. This will show the pitch of the aircraft as indicated by the relative height of the aircraft above the horizon vector line, and the rate of turn indicated by the angle of the horizon line which will slope towards either the left or right during a turn. Hence the pitch and the rate of turn are shown on a common instrument.

To the upper right of the display are two main navigation instruments, the compass showing the heading of the aircraft from 0° due north through 180° south, 270° west etc. Both the instrument needle and the displayed numeric value indicate the direction the aircraft is heading, but not taking into account the effects of the prevailing wind. The bearing indicator to the right

SECTION 1:3 Cont'd.

shows the actual direction to be flown in order to reach the destination runway.

Below these is the VSI or vertical speed indicator which shows the rate of climb or descent in feet per minute. If climbing the value will be positive and if descending the value will be negative. During level flight the value will be close to zero. Immediately below this is the altitude value showing the height of the aircraft in feet above sea level, which for the Simulator is also taken to be ground level at all airports.

Two instruments remain. The first shows the IAS or indicated air speed of the aircraft. The stall speed, which will vary with load, flap settings etc, is also displayed on this instrument, along with a strip indicator giving an indication at a glance if the airspeed is safely above the stall speed when coloured green, above but close to the stall speed if blue, or below stall if red. The airspeed as indicated can only be taken as an approximate value, particularly at higher altitudes. Speed relative to the ground may be greater by a considerable margin depending on the strength and direction of the wind, whether it be a strong head wind opposing the motion of the aircraft or a tail wind. At higher altitudes with reduced air density, the air pressure acting on sensors or transducers for the speed indicator is also reduced giving an artificially low speed value.

The final instrument shows the engine throttle setting and the effective power output from the engines. The throttle setting may be any value from full power, through zero, to full reverse thrust. The output power is shown as a percentage value, depending on the number of engines running of course, the throttle setting, and the altitude effect which will reduce the true power output with reducing air density and available oxygen supply to the engines as the aircraft climbs.

1:4 START-UP PROCEDURE

We left section 1:2 with the trip selected and the aircraft fuelled. The next stage is the start-up procedure prior to take-off. On the Flight Simulator this simply involves starting the four engines and setting the flaps down to provide the required lift at take-off speed.

Only two engines may be started up or shutting down at one time. Hence two engines will only respond and start up once the first two are fully running. The four engines are either started up or shut down with the '1' to '4' keys respectively.

For take-off the flaps should be lowered 20° which will provide adequate extra lift even with a full load of 150 tonnes of fuel on board.

1:5 TAKE-OFF

With all engines run up and the flaps set, apply full throttle with the brakes still applied. When 100% or full power is applied, release the brakes. The flight clock will start and the speed of the aircraft will start to increase.

The effect of the cross wind may cause the aircraft to drift either left or right across the runway. Use the aileron controls to compensate and correct for this, normally 1° of correction in the heading as indicated on the compass will be sufficient. Further correction may be required as the aircraft speed increases and before the craft climbs off the runway.

SECTION 1:5 Cont'd.

The runway display will show the length of runway used, 3 miles being the total available. Once the air speed is above the stall speed, with the IAS bar indicator shown green, set the elevators to a value of +8 or +9. The pitch of the aircraft will increase and once this reaches 8° or 9°, depending on the speed, fuel load etc, the aircraft will leave the ground. The VSI and altitude instruments will show the rate of climb and altitude.

If the attitude or pitch of the aircraft on the ground exceeds +10° the tail of the craft will hit the ground resulting in a crash. Hence it is necessary to ensure the speed of the aircraft is sufficient to provide the required lift at a pitch not exceeding this value. At the same time it is equally important to have the aircraft airborne before the available length of runway is exceeded or the results will be just as severe.

Immediately the aircraft is in the air, retract the landing gear and reset the elevators to the zero position. Depending on the fuel load, reduce the throttle to give 70% to 80% power output. The speed will continue to increase and the rate of climb also. The flaps should be raised in stages to fully raised or zero setting as the speed increases.

Use the aileron controls to turn the aircraft as required to fly towards the destination airport. Note that during a turn the nose of the aircraft will tend to drop and the rate of climb will reduce. The more severe the turn, the more pronounced the loss of pitch. A slight increase in the setting of the elevators will be required to compensate for this.

As the aircraft continues to accelerate, the rate of climb will increase. The VSI should normally be maintained at 2500 to 3000 ft/min by using the elevator controls to adjust the pitch of the aircraft. If necessary also adjust the throttle to keep the speed of the aircraft within the cruise range of 375 to 410 knots. Once the aircraft reaches the required cruise altitude, reduce the pitch and if necessary the throttle slightly, until the VSI indicates level flight and the IAS is steady.

1:6 FLYING AND NAVIGATING

At this stage we have reached the most straight-forward part of the flight and until we start to descend and approach the destination airport, the control of the aircraft will require only minor adjustments.

As the flight progresses and fuel is consumed reducing the overall load, speed will increase slightly and the aircraft will start to climb gradually. The wind strength and in particular the wind direction may also vary considerably. Hence it will be necessary to adjust the pitch, heading, and throttle to maintain level flight in the required direction without exceeding the maximum safe speed.

At any time after take-off the map display may be used to give a summary of flight conditions, including the destination runway bearing for those who neglected to notice it during the flight selection process. The map will also show the positions of the originating and destination airports and the present aircraft position. Refer to section 2:1 for further details on the map display.

The Time Skip feature may also be used during this phase of the flight. Refer to section 2:4 for full details of this option.

SECTION 1:6 Cont'd

Before entering the range of the ILS or instrument landing system, it is necessary to make the initial approach and descent to ensure the aircraft will be on the correct course for the required destination runway bearing and at approximately the correct altitude for the ILS guidance.

Dealing firstly with the descent of the aircraft. The optimum glideslope or descent path is 250 feet per mile. One or two examples will illustrate this quite simply. At a distance of 100 miles from the destination the altitude on approach should be $100 \times 250 = 25000\text{ft}$. At a distance of 25 miles which is the maximum range of the ILS system the altitude should be $25 \times 250 = 6250\text{ft}$. The start of the descent will depend on the cruise altitude, the higher the altitude, the earlier descent will need to commence. If the descent is started later then the required rate of descent will be correspondingly greater. The main requirement will still to be at about 6250ft to intercept the ILS beam at the maximum range of 25 miles.

The required rate of descent in ft/min to maintain this glideslope will vary according to the true ground speed, with approximate values as follows; at 360knots -1500ft/min; 300knots -1250 ft/min; 240knots -1000ft/min; and 180knots -750ft/min. These values are based on true ground speed which will vary from the IAS and it will therefore be necessary to mentally check and compare the altitude and distance-remaining values.

To descend under full control reduce the throttle and decrease the pitch of the aircraft. As the speed of the aircraft reduces the rate of descent will increase. To control this adjust the throttle and pitch accordingly while allowing the speed of the aircraft to reduce gradually. To ensure full control as the speed reduces the flaps should be lowered by 10° at approximately 280 to 300 knots. The extra drag caused by the flaps will in itself cause the speed to reduce further and if necessary the throttle may be adjusted to compensate for this. When approaching the range of the ILS the flaps should be lowered to 20° , the speed adjusted to 180 to 200 knots depending on fuel load and the pitch adjusted to give a VSI of -780 to -840ft/min. Speeds much below this may make control difficult while excessive speed will present difficulty in landing and stopping the aircraft within the confines of the runway.

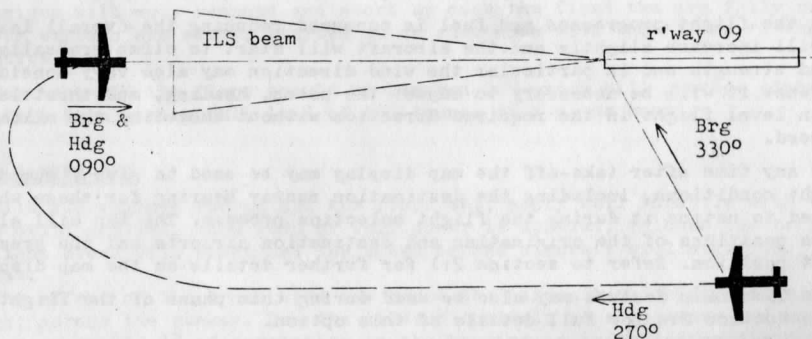


Figure 3

SECTION 1:6 Cont'd.

The second consideration during the initial approach is that of the direction required to intercept the ILS glidepath. Figure 3 will illustrate the requirement to align the aircraft with the designated runway.

Initially the bearing to the airport, as shown on the instrument display, may be quite different from the required runway bearing. It is necessary to bring these two to coincide.

Figure 3 initially shows the aircraft flying west at 270° heading, while the bearing to the airport is 330° . Hence the aircraft will continue west, following the dotted flight path, and then turn right to approach with a bearing of approximately 090° which coincides with the runway 09 (090°).

It is much easier to control the approach making a relatively wide sweep to intercept the ILS beam as far out as possible, and then stabilise the rate of descent and the heading within the scope of the beam. Navigating and intercepting the beam close in to the runway will provide difficulty in stabilising the approach while remaining within the scope of the ILS.

1:7 INSTRUMENT APPROACH - ILS

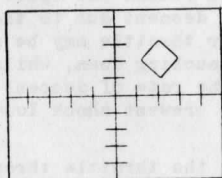
The instrument landing system will provide a visual instrument display of both the glideslope and glidepath during approach, provided the aircraft has intercepted and remains within the ILS beam as outlined in the previous section. The principal instrument is at the top centre of the display.

While in the range and scope of the beam the position of the aircraft will be indicated on this grid, showing both the variation in altitude from the ideal glideslope and the directional or bearing drift on the horizontal scale.

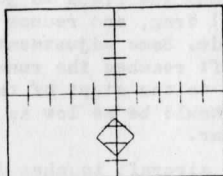
Three markers are associated with the ILS system. The upper or ILS marker will show green if the aircraft is accurately positioned and following the ILS approach, yellow if the aircraft is outside but close to the scope of the ILS, or red if the approach is incorrect. It is important the marker be in the green state indicating accuracy during the final approach or the landing will have to be abandoned and the approach repeated.

Correction to both the rate of descent and heading with adjustments will be required during the instrument approach, particularly to compensate for variations in wind strength and direction. The grid indicator should be kept at the centre, or as close as possible, to the centre of the cross-hairs.

Figure 4 shows two examples of the indicator on the ILS grid.



4a



4b

Figure 4

SECTION 1:7 Cont'd.

In figure 4a the grid cross-hairs are shown below and to the left of the marker. Hence the aircraft is low and to the left of the beam centre. To compensate, reduce the rate of descent slightly and turn right 1 or 2 degrees. Do not overcorrect or it may be difficult to stabilise once the cross-hairs are intercepted.

In figure 4b the marker is below the grid centre, hence the aircraft is too high. Increase the rate of descent slightly to correct for this.

The simple rule to avoid confusion is to adjust the VSI or heading or both to take the cross-hairs to the marker, not the other way around.

During the instrument approach the speed of the aircraft should still be maintained at 180 to 200 knots with 20° flaps down and with the pitch at 2° to 4° to give a VSI of -720 to -840 ft/min. The power required should at this stage be 25% to 30%.

At a distance of 10 miles from the runway the outer marker will flash to advise the distance remaining is within 10 miles. Later during the final or visual approach the inner marker IM will flash to warn that the aircraft is only 1 mile from the runway threshold.

One important step, but one which is very easily neglected, during the approach is to lower the landing gear. This must only be done once the IAS has fallen below 240 knots or the drag on the gear will be excessive, causing damage and making a safe landing impossible. Lowering the landing gear, with the additional drag will require a slight increase in power to maintain the speed and stable descent.

If the correct approach is completed, the ILS will operate until the aircraft is 4 miles from the runway. At this point, provided the ILS marker is green, the runway will become visible. If this does not occur the aircraft must abandon the approach, circle and repeat the approach.

1:8 FINAL APPROACH AND LANDING

If the instrument approach has been successful and the runway is now displayed, the relative altitude and side drift of the aircraft will be shown. The distance required to reach the runway (-ve runway) or the length of runway used (+ve runway) are also shown.

During this final approach the rate of descent should be maintained and the side drift corrected with the aileron controls to keep the aircraft above the runway.

When the inner marker flashes indicating the aircraft is 1 mile from the runway threshold, lower the flaps to 30°. This will reduce the speed gradually due to the additional drag, and reduce the rate of descent due to the extra lift the flaps provide. Some adjustment to pitch or throttle may be required to ensure the aircraft reaches the runway before touching down, while landing as close as possible to the start of the runway. The rate of descent at the point of touchdown should be as low as possible to prevent shock loadings from damaging the gear.

Immediately the aircraft touches down, reduce the throttle through zero to full reverse thrust to decelerate the aircraft. Adjustment to the heading may still be required to ensure the aircraft does not drift off the side of the runway.

SECTION 1:8 Cont'd.

The wheel brakes cannot be applied until the indicated speed is no more than 40 knots, by which time the reverse thrust can be progressively reduced. Once the brakes are applied the throttle may be reset to zero. If all is correctly controlled the aircraft will come to rest within the boundaries of the runway.

It must be appreciated that the final approach and landing is the most difficult part of the flight and for newer users particularly some practice may be required before reasonable success is achieved. The Practice Landing routine outlined in section 2:5 will provide assistance to those consistently experiencing difficulty.

1:9 SHUT-DOWN PROCEDURE

All that remains to complete the flight is to shut down the engines and adjust all control surfaces to their zero positions. As for start-up, the four engines may only be shut down two at a time using the same control keys. The throttle must be set to zero if this has not already been done, the flaps fully raised and the aileron and elevator controls set to the central or zero position. The brakes must be left applied and of course the landing gear left locked down.

On completion of these steps the flight clock will stop and the display will acknowledge the safe completion of the flight and prompt the user to select or decline a further flight. The user may now feel well satisfied at having flown on the Flight Simulator through a full flight.

SECTION 2

This section deals with the various options available within the programme in some detail. Each option has it's own series of commands and conditions under which it may be used.

2:1 MAP DISPLAY

When a flight has been selected and the actual flight has commenced, the map display remains available to the user by the use of the 'M' command. Displayed alongside the map will be a summary of current flight information which includes distance remaining to reach the destination, bearing to the destination, current wind details, aircraft altitude and heading and speed relative to the ground, and just in case the user neglected to make careful note of it in the flight selection process, the required landing runway at the destination airport. The map itself will display the position of the departure airport (non-flashing), the destination airport (slow-flashing), and the current aircraft position (rapid-flashing).

While the map and flight summaries are displayed the flight clock is stopped and the actual flight suspended. Hence the map may be displayed for as long as the user desires, without any risk of the aircraft crashing or even moving. If the user is called away or simply wishes to take a break, the map display is one convenient method of totally suspending the programme while preserving all flight data.

The 'M' command to select the map display will not function while the runway is displayed on the flight instrument display, either during take-off or final approach and landing, or while a warning message is being displayed on the screen.

To return from the map display to the full flight instrument display and continue the flight, press the 'ENTER' key.

While in this section we have mentioned one method of suspending the programme, one or two other small matters should be clarified. Firstly the 'STOP' key. If this is pressed on it's own, this will also effectively suspend the programme quite safely, even if the flight instrument panel is displayed and the clock running. Pressing the 'STOP' key again will return the programme to running normally.

Secondly, the 'CTRL=STOP' function will have no effect and is disregarded by the programme. The user-definable function keys are disabled at the start of the programme and will only be reinstated when the user declines the prompt for a further flight as outlined in Section 1.

2:2 ABANDON FLIGHT

At any time after the flight selection procedure has been completed the flight may be abandoned by pressing the 'ESC' key. This will terminate the flight and display on the screen the prompt for a further flight as detailed in Section 1. The user may select a further flight or end the run of the programme as previously described.

2:3 DESTINATION DIVERT

If after take-off, it is desired to shorten a flight by landing at an airport which may be closer than that originally selected, or if the fuel remaining appears inadequate to reach the destination safely, this programme option will allow the selection of the alternative airport.

This option will not function during the flight selection procedure, or while using the practice landing routine, or while the runway is displayed on the full instrument display during take-off, final approach or landing in a normal flight.

To select a new destination, firstly display the map and current flight summary with the 'M' command. The Divert routine can now be brought into effect with the 'D' command which will modify the flight information summary and show the distance, bearing, and runway for an alternative destination. The fuel available is also displayed to allow the user to check approximately if the alternative destination would be within flying range. The map itself will indicate the current position of the aircraft and the locations of both the originally selected and alternative destinations.

Three options are now available to the user and each of these is detailed separately as follows:-

- a) To select another airport as possible destination simply use the 'D' command again and the details and map will be revised accordingly. This step may be repeated as many times as required, with all airports eventually being offered as alternative destinations, the only exception being the originally selected destination.
- b) To confirm any particular airport as being the required destination, with the details of that airport on the display, press the 'ENTER' key. This will modify the programme data and display the normal full current flight summary alongside the map, with the new airport as the destination. To continue the flight to the new destination, press the 'ENTER' key once again and the full flight instrument display will return to the screen, the clock will restart and the flight will continue.
- c) To cancel the Divert routine and continue to the original destination, use the 'C' command. The normal flight summary will again be displayed and pressing the 'ENTER' key will subsequently return the programme to the flight instrument display and resume the flight.

2:4 TIME SKIP

During a flight the user may reduce the actual time required to complete the flight with the 'T' command. Each time the Time Skip functions the flight clock is advanced 10 minutes and the aircraft position, fuel remaining, distance remaining and true bearing to the destination are all revised and the relevant sections of the display updated.

If the requirements as listed below are met, the Time Skip will interrupt the flight and once the display is updated, the programme will await further input.

Pressing the 'ENTER' key will restart the clock and resume the normal flight mode. Alternatively the 'T' command may be used again and, providing the same requirements are still met, the Time Skip will function once again and await further input as before.

SECTION 2:4 Cont'd.

If the Time Skip fails to function or fails to repeat after functioning it will be necessary to resume normal flight with the 'ENTER' key and, if appropriate, adjust the aircraft attitude, heading, etc so that the requirements are met, before the feature may be used effectively again.

The conditions which must be complied with for the Time Skip to function with the 'T' command are as follows:-

- 1) The aircraft must be in relatively level flight with a rate of climb or descent not exceeding 60fpm.
- 2) The speed of the aircraft must be steady.
- 3) The aircraft elevator and aileron controls must be in their central or zero positions.
- 4) The aircraft must be flying towards the destination airport with the compass heading within 10° of the required bearing as shown on the display.
- 5) The distance remaining to reach the destination must be at least 150 miles.
- 6) There must be a minimum of 6 Tonnes of fuel remaining.

2:5 PRACTICE LANDING

For users having difficulty in achieving successful instrument approach and final landing and shutdown within the confines of the runway, this routine will be of assistance. The Practice Landing routine is selected with the 'P' command at the initial stage of the flight selection procedure, but will not function if any other departure or destination airport has already been selected.

This routine will preset all relevant programme data to set the aircraft 30 miles out from and approaching Auckland airport, without the user being required to complete the usual flight selection, take-off and full flight to reach this position and attitude. Wind effects, while not being completely eliminated, are minimal to make the approach and landing a little more straightforward than in a normal full flight.

After the 'P' command functions the normal current flight summary will be shown, and subsequently pressing the 'ENTER' key will display the full flight instrument panel, start the clock and the flight will proceed. All controls and flight parameters, apart from the reduced wind effects are as detailed in Section 1 of this guide.

SECTION 3

The following provides a reference guide to the controls and relevant programme criteria which should be observed for the Simulator to be used fully.

3:1 CONTROLS

These are summarised in groups in accordance with the respective parts of the programme in which they are effective, these groups including both aircraft controls and programme inputs.

Additionally, several primary aircraft controls are available for the use of either one or two joysticks, including their respective trigger buttons. Joystick controls are summarised separately below. If using joysticks which have auto-fire buttons fitted, ensure these buttons are switched to the 'off' position.

AIRCRAFT CONTROLS

'1' to '4'	Start-up / Shut-down respective engines.
'B'	Apply / Release brakes.
'G'	Lower / Raise landing gear.
'F'	Lower flaps 10°
'R'	Raise flaps 10°
'+'	Increase throttle
'-'	Decrease throttle

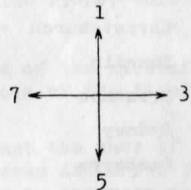
JOYSTICK CONTROLS

JOYSTICK 1

Forward (1)	Elevator down - decrease pitch.
Back (5)	Elevator up - increase pitch.
Right (3)	Aileron/Rudder Turn right - to starboard.
Left (7)	Aileron/Rudder Turn left - to port.
Trigger	Raise / Lower landing gear.

JOYSTICK 2

Forward (1)	Increase throttle.
Back (5)	Decrease throttle.
Right (3)	Lower flaps 10°.
Left (7)	Raise flaps 10°.
Trigger	Apply / Release brakes.



PROGRAMME CONTROLS

'STOP'	Suspend programme execution. Press again to resume programme.
'ESC'	Abandon current flight.
'Y' or 'N'	Confirm or decline prompt for further flight.

SECTION 3:1 Cont'd.

FLIGHT SELECTION

'P'	Select practice landing routine.
' ' or ' '	Select airports from list.
'ENTER'	Confirm destination/departure airports.
'0' to '9'	Select fuel load.
'ENTER'	Confirm fuel load to programme data.
'ENTER'	Proceed to full flight instrument display.

MAP DISPLAY

'M'	Select map display.
'ENTER'	Return to flight mode.

DIVERT ROUTINE

'M'	Select map display.
'D'	Select alternative destination, repeat as required.
'C'	Cancel routine.
'ENTER'	Confirm new destination.
'ENTER'	Return to flight mode.

TIME SKIP ROUTINE

'T'	Skip ten minutes, repeat as required.
'ENTER'	Return to flight mode.

3:2 AIRPORT RUNWAYS

The following list the possible runway bearings at all airports for take-off and landing. The optional direction will depend on the prevailing wind during the actual flight.

Auckland	05 / 23	bearing ⁰ 050 / 230
Wellington	16 / 34	160 / 340
Christchurch	02 / 20	020 / 200
Dunedin	03 / 21	030 / 210
Brisbane	04 / 22	040 / 220
Sydney	16 / 34	160 / 340
Canberra	17 / 35	170 / 350
Hobart	12 / 30	120 / 300
Melbourne	16 / 34	160 / 340
Adelaide	18 / 36	180 / 360
Alice Springs	12 / 30	120 / 300
Perth	02 / 20	020 / 200
Darwin	11 / 29	110 / 290

SECTION 3 Cont'd.

3:3 PERFORMANCE NOTES

These details indicate various performance and programme criteria as a guide for users to help prevent unnecessary failures during attempted flights on the Simulator.

Fuel Load	10-150 Tonnes.
Cruise Speed	375-410 Knots.
Maximum Safe Speed	415 Knots.
Absolute Maximum Speed	425 Knots.
Maximum Gear-down Speed	240 Knots.
Cruise Altitude	10000-40000 Ft.
Maximum Safe Altitude	44000 Ft.
Absolute Maximum Altitude	45000 Ft.
ILS Range	25 Miles.
ILS Glideslope	250 Ft per Mile.
Runway Visible From	4 Miles.
Maximum Safe Descent	
Rate on Touchdown	-600 Ft per Minute.
Runway Length	3 Miles.

3:4 WARNING MESSAGES

Under some circumstances there may be imminent danger of crashing the aircraft or overstressing structural parts of the aircraft. In such cases a warning message will be displayed on the flight instrument panel conspicuously in that segment where flight wind details are normally shown. These prompts will generally give the user sufficient warning to allow corrective measures to be taken. However, this is not always the case and it is possible the aircraft may be unable to complete the flight safely. The list below summarises the messages and conditions under which they are displayed.

TOO LOW - PULL UP

If at any time apart from during the final approach, the aircraft is descending and the altitude falls below 750ft, this message will be displayed.

**STALLING - APPLY
POWER - PULL UP**

Displayed if the speed of the aircraft falls to within 20 knots of the displayed stall speed.

**TOO HIGH - STRUCTURAL
RISK**

This message will prompt the user if the aircraft altitude exceeds 44000 ft. If the altitude continues to increase and exceeds 45000 ft then structural failure will occur and the aircraft will crash.

**TOO FAST - STRUCTURAL
RISK**

This warning will be conveyed if the IAS of the aircraft reaches a value of 415 knots. If power is not reduced and the speed continues to increase and reaches 425 knots then structural failure will occur and the aircraft will crash.

SECTION 3:4 Cont'd.

LOW FUEL RESERVE

This warning will be displayed briefly to advise the user that the remaining fuel reserve has fallen below 6 tonnes.

TOO FAST - LANDING
GEAR DAMAGED

The landing gear will be damaged if not raised on take-off before the speed of the aircraft reaches 240 knots, or if the landing gear is lowered at any time while the speed exceeds this value. The landing gear may still be raised or lowered but will not support the aircraft and will collapse on landing.

Additional to these messages are those which indicate the relative quality or standard of the landing, assuming the landing was not sufficiently heavy to collapse the landing gear, the acknowledgement of a safely completed flight on completion of the shut-down procedure, and the prompt for a further flight.